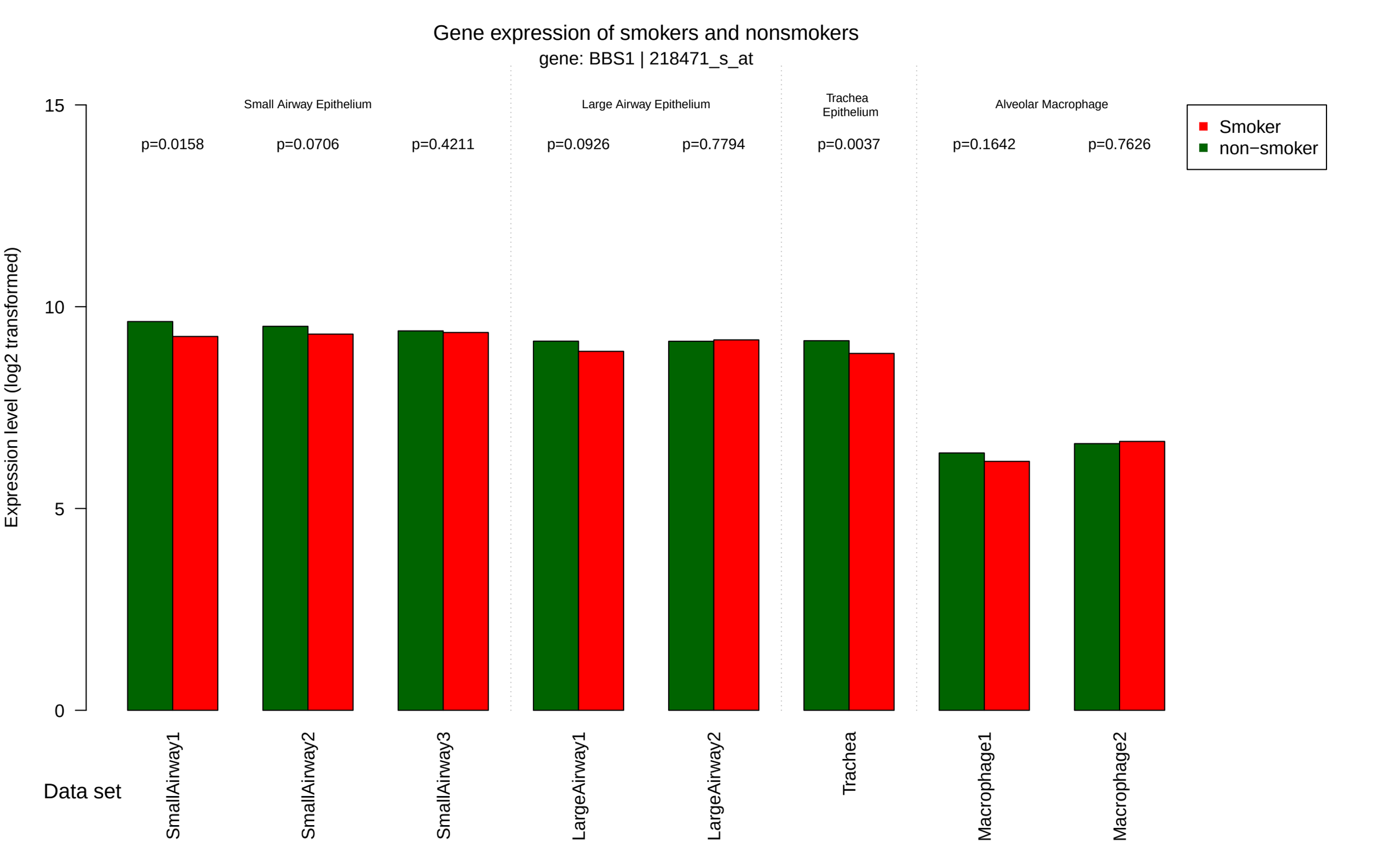
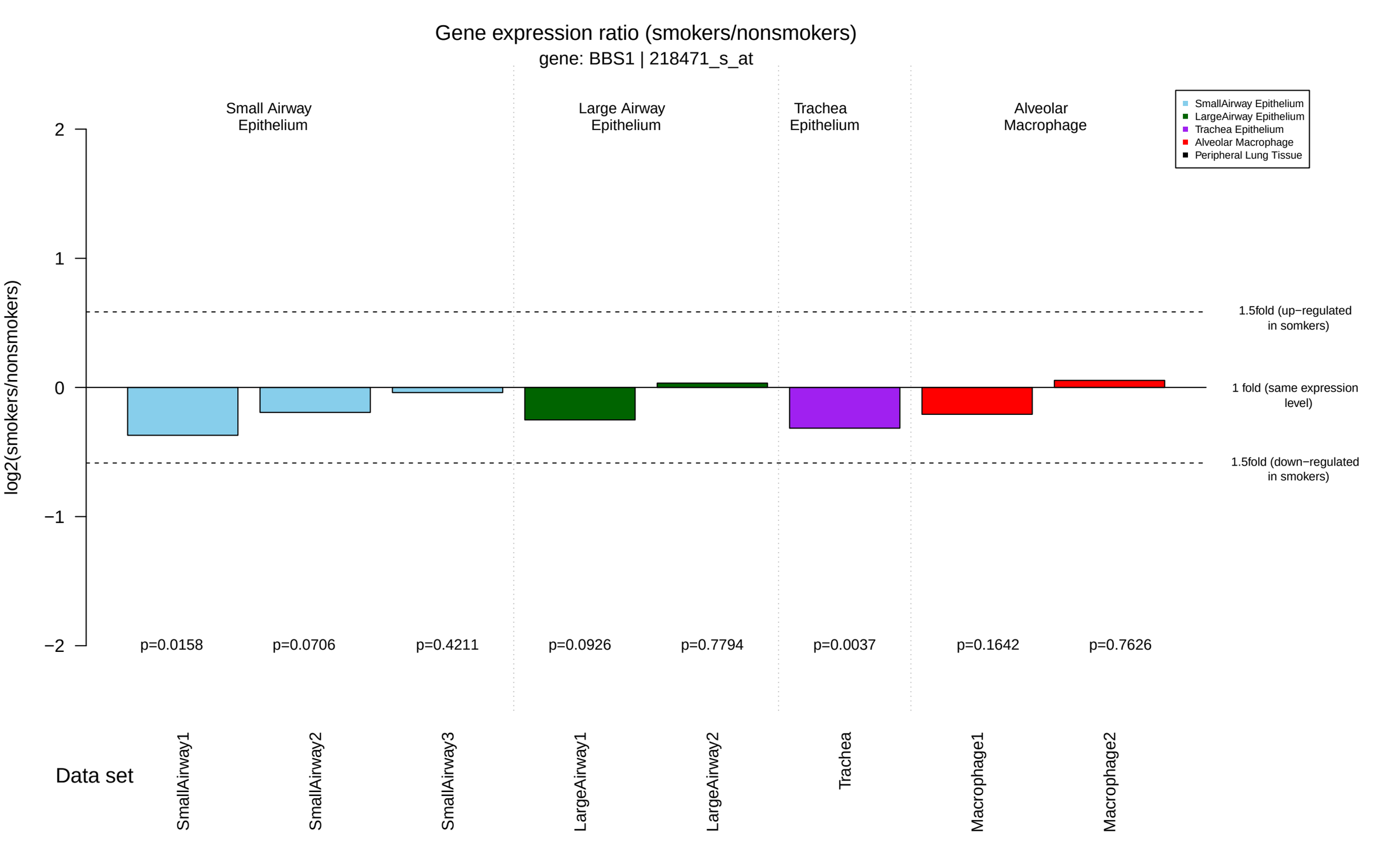
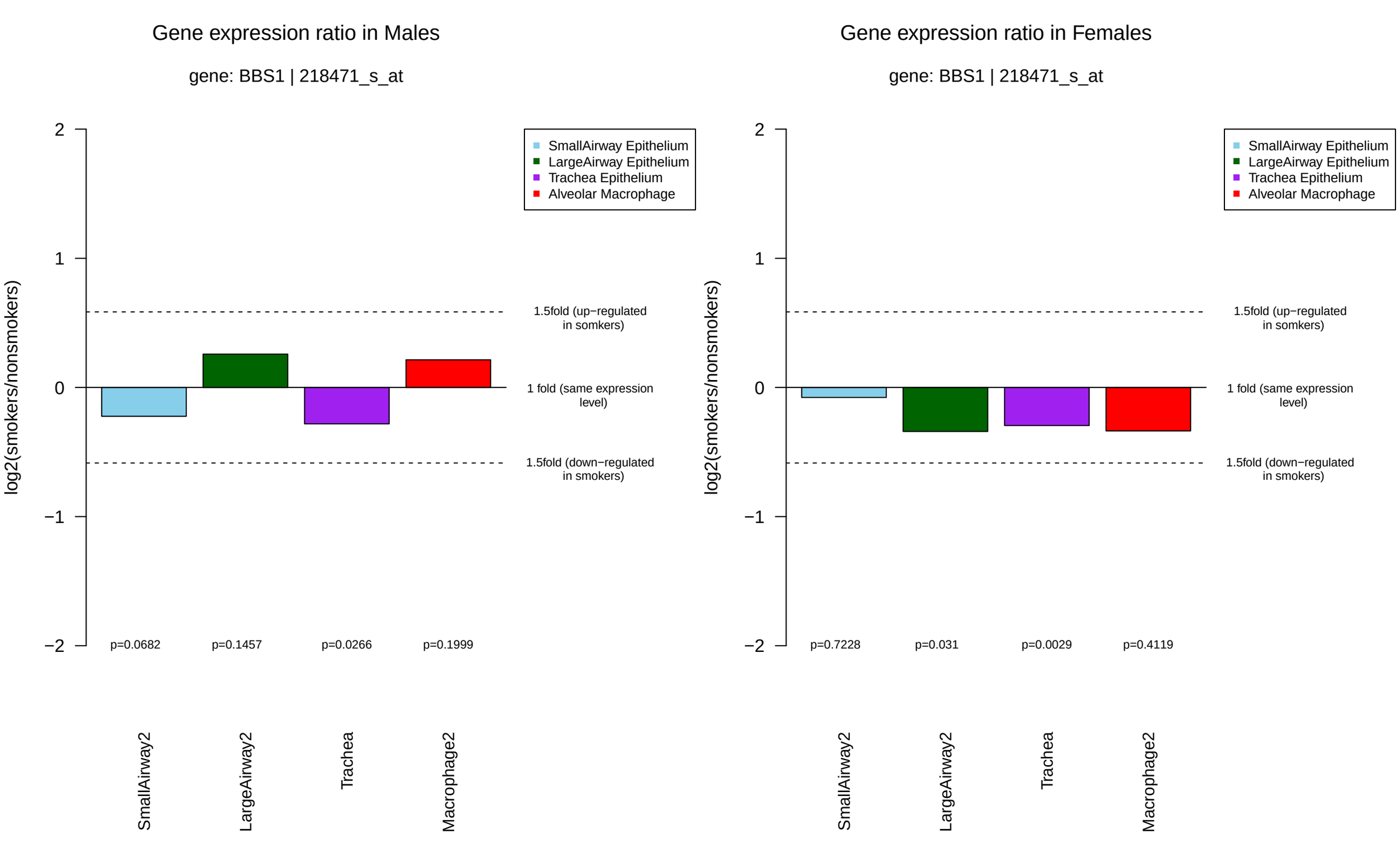
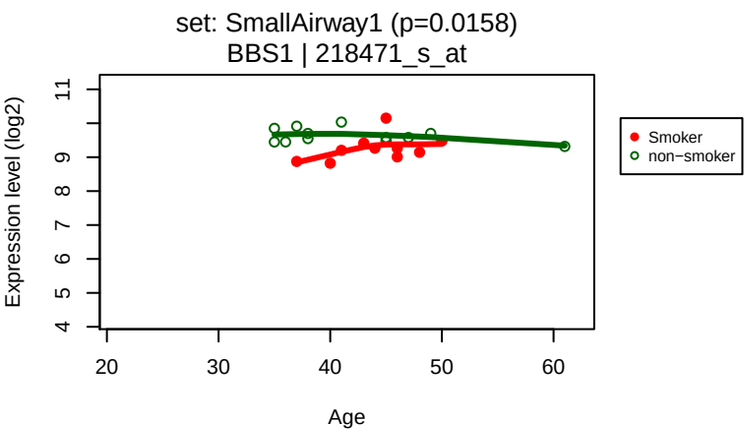
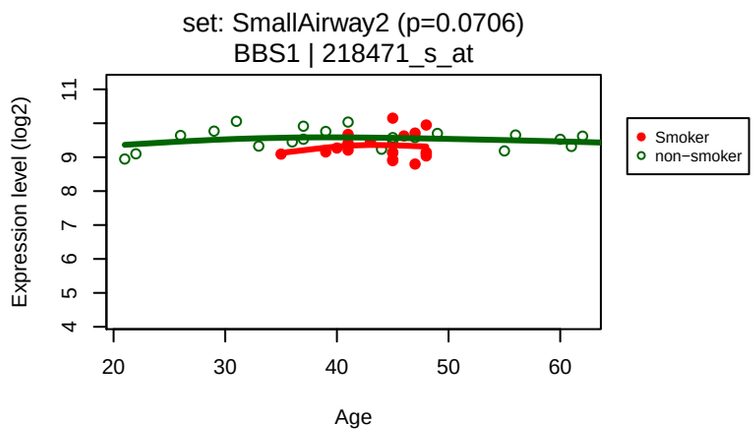
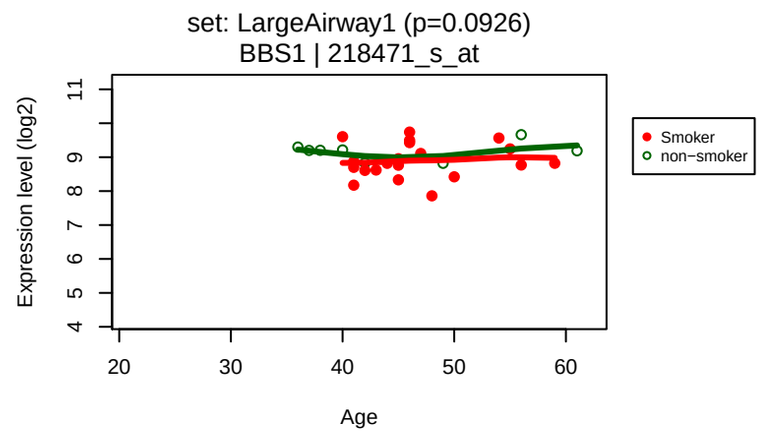

Gene expression of smokers and nonsmokers
gene: BBS1 | 218471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0158
p=0.0706
p=0.4211
p=0.0926
p=0.7794
p=0.0037
p=0.1642
p=0.7626
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

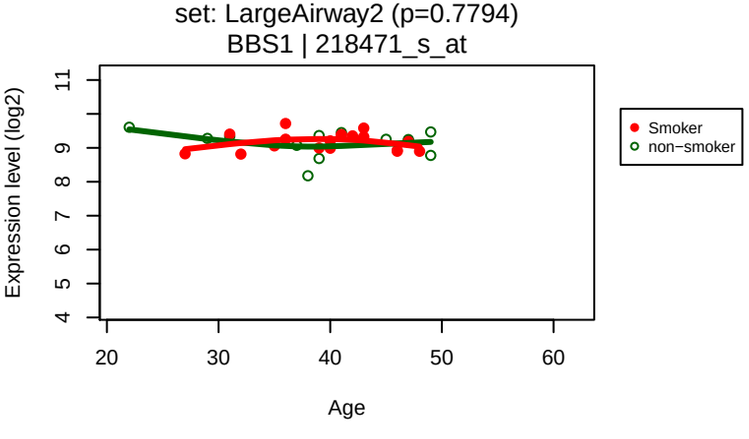

Gene expression ratio (smokers/nonsmokers)
gene: BBS1 | 218471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0158
p=0.0706
p=0.4211
p=0.0926
p=0.7794
p=0.0037
p=0.1642
p=0.7626
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

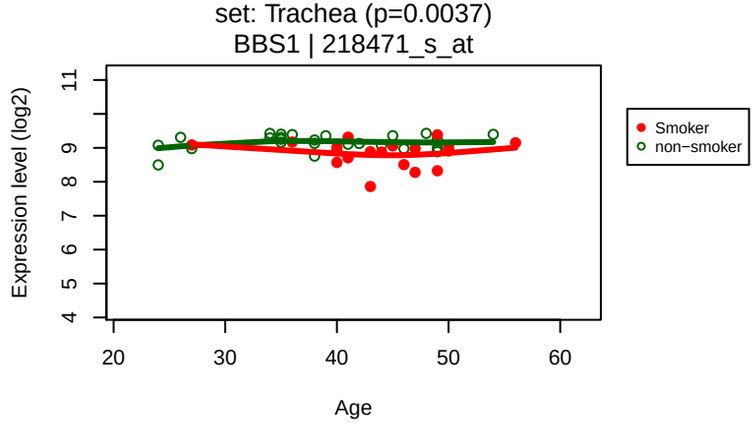

Gene expression ratio in Males
gene: BBS1 | 218471_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0682
p=0.1457
p=0.0266
p=0.1999
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BBS1 | 218471_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7228
p=0.031
p=0.0029
p=0.4119
SmallAirway2
LargeAirway2
Trachea
Macrophage2

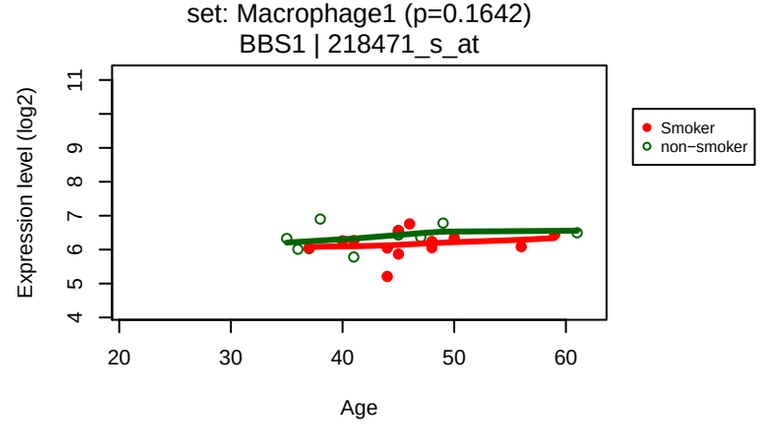

set: SmallAirway1 (p=0.0158)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker

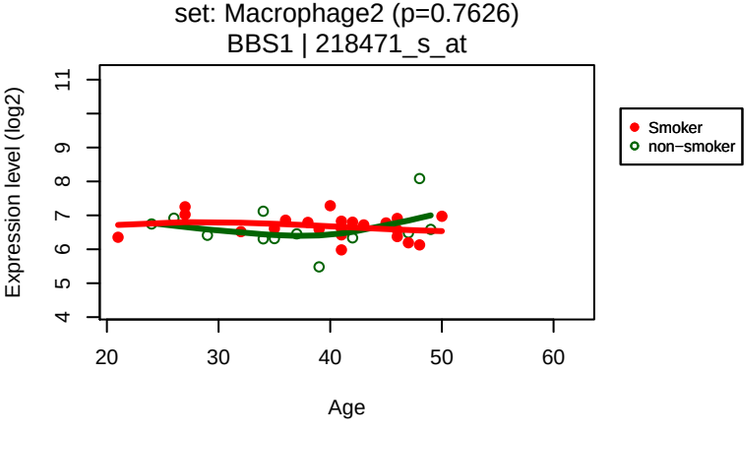

set: SmallAirway2 (p=0.0706)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker

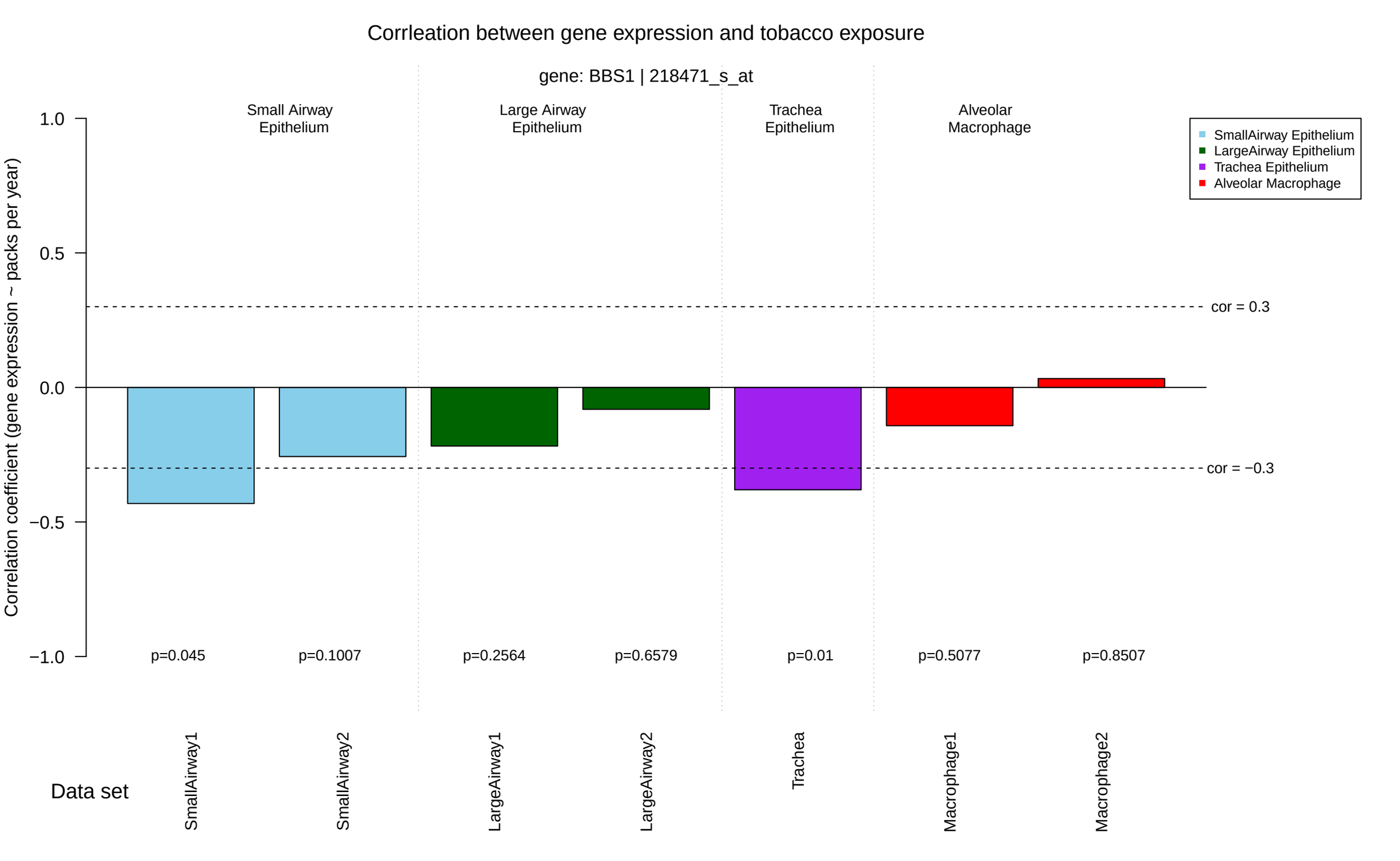

set: LargeAirway1 (p=0.0926)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7794)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1642)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7626)
BBS1 | 218471_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BBS1 | 218471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.045
p=0.1007
p=0.2564
p=0.6579
p=0.01
p=0.5077
p=0.8507
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2